

Harshita Vachhani

✉ harshitavachhani336@gmail.com ☎ +91-7231067620 in Harshita Vachhani 🌐 Harshita Vachhani 🌐 Portfolio

Education

Indian Institute of Technology, Jodhpur
B.Tech in Electrical Engineering

Aug 2023 – May 2027
CGPA: 8.54/10.0 (till 6th semester)

St. Anselm's Senior Secondary School, Jaipur
Central Board of Secondary Education

July 2010 – Apr 2023
Class 12: 95.6% | Class 10th: 96.6%

Projects

Image Retrieval

🔗 [GitHub](#)

- Developed a robust image retrieval system leveraging classical machine learning algorithms on the CIFAR-10 dataset to efficiently identify and retrieve visually similar images.
- Implemented a classification pipeline to categorize input images and performed feature-based similarity search within the predicted class.
- Achieved over **91%** accuracy and built a user-friendly web interface using streamlit to allow users to upload images and view retrieval results.
- Tools & Technologies:** Python, scikit-learn, Streamlit, Matplotlib, Torch, Torchvision, Joblib, Git/GitHub.

Typing Speed Test Game

🖥️ [Live Demo](#) 🔗 [GitHub](#)

- Created an interactive typing speed test web application to help users improve typing skills by providing real-time performance metrics.
- Implemented live WPM (Words Per Minute) calculation, and accuracy tracking, which updates dynamically as the user types.
- Tools & Technologies:** HTML, CSS, JavaScript, Git/GitHub.

Blockchain Consensus Protocol

🔗 [GitHub](#)

- Designed and implemented a election based blockchain using Directed Acyclic Graph (DAG) to enable parallel block creation and improve transaction throughput.
- Implemented a validator-weighted fork selection algorithm (GHOST) to choose the main chain fairly.
- Implemented Fruitchain reward mechanism to improve network participation.
- Added a governance system to manage validators, incentivize participation, and finalize blocks securely.
- Tools & Technologies:** Solidity, Ganache, MetaMask, HTML, CSS, Javascript, Git/GitHub

Technical Skills

- Programming:** C/C++, Python, MySQL
- Tools and OS:** Arduino IDE, Figma, MATLAB, Git/GitHub, Google Colab, Linux, Windows
- Web Development:** HTML, CSS, JavaScript, React, Express
- Libraries and Frameworks:** Pandas, NumPy, scikit-learn, PyTorch, Matplotlib, TensorFlow, Streamlit, Gradio, OpenCV
- Core Competencies:** Data Structures and Algorithms, Object Oriented Programming, Operating Systems, Database Management System, Computer Networks

Coursework

- Data Structures and Algorithms, Computer Organization and Architecture (**A-**), Pattern Recognition and Machine Learning (**A-**), Deep Learning
- Linear Algebra, Calculus, Probability Statistics and Stochastic Process(**A**)
- Signals and Systems(**A**), Analog Circuits (**A-**), Control Systems (**A-**), Communication Systems (**A-**), Data Communication Networks (**A**), Embedded Systems

Achievements

- Achieved 6th Department Rank among 90+ students at the end of 6th semester.
- Secured AIR 10548 in JEE Advanced 2023 among 1.8 Lakh shortlisted candidates.
- Solved 500+ problems on Leetcode with a rating of 1654. Profile: [Harshita](#) 

Positions of Responsibility

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| ◦ Internship Representative Office of Training & Placement, IIT Jodhpur | May 2025 - Apr 2026 |
| ◦ Support Team Member Office of Training & Placement, IIT Jodhpur | Oct 2024 - May 2025 |
| ◦ Assistant Head Prometeo '25, IIT Jodhpur | Nov 2024 - Jan 2025 |